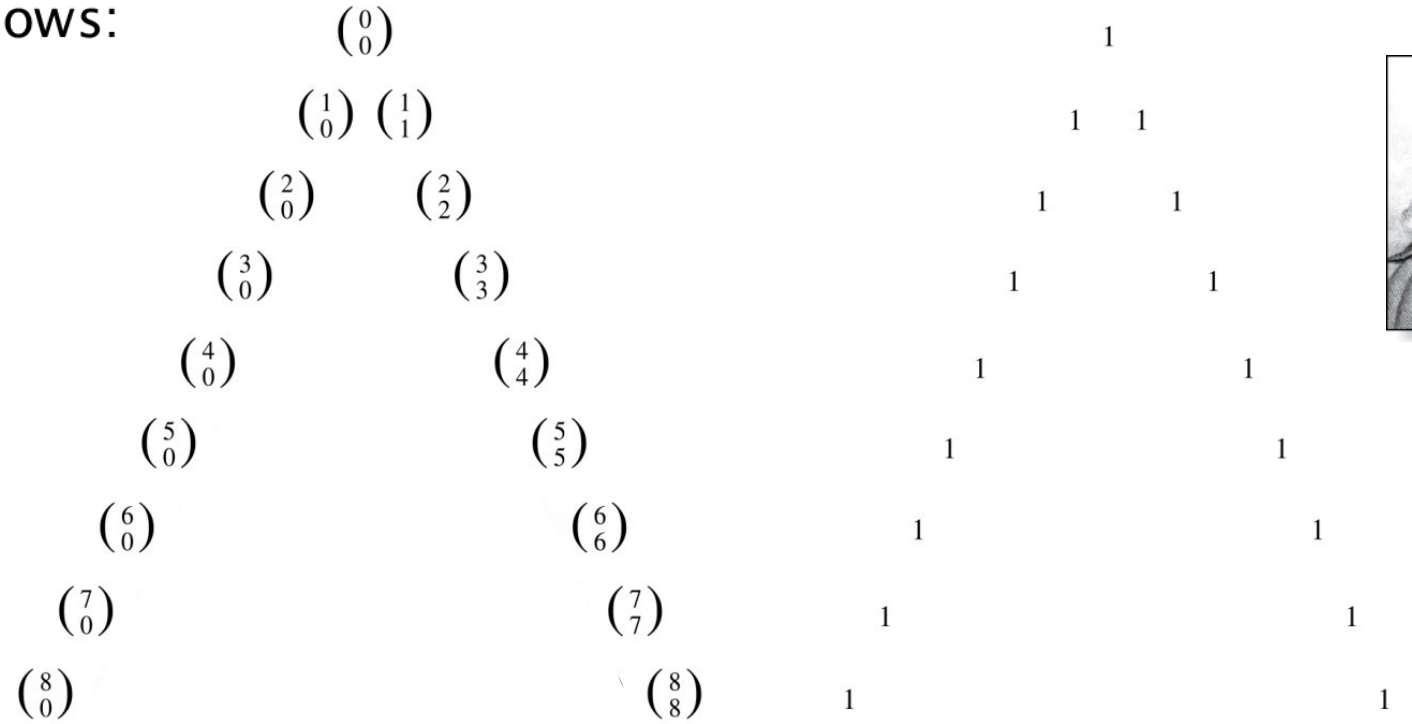
Pascal's Triangle (1641)

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

- **Corollary**

Pascal's identity, together with the initial conditions $\binom{n}{0} = \binom{n}{n} = 1$ for all integers *n*, can be used to give a geometric arrangement of the binomial coefficients in a triangle, called Pascal's triangle, as follows:



(a) (b)

